

AYMANE AIT DADS

Bioinformatics & Data Science Intern | AI/ML for Biological Data · Computational Analysis · Predictive Modeling

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PROFESSIONAL SUMMARY

Data Science engineering student at EURECOM who built and deployed Random Forest and K-Means models on 100,000+ real-world network records at Orange Maroc, reducing manual analysis time by approximately 60% and directly informing network optimization decisions. Ranked 17th out of 3,000+ teams (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a 30B-parameter model with LoRA SFT, demonstrating the ability to apply advanced AI/ML methods to structured, multi-category data problems. Brings hands-on experience in Python, Pandas, Scikit-learn, feature engineering, clustering, ensemble methods, and data visualization with Power BI - tools directly aligned with computational and AI/ML-driven drug discovery workflows. Aims to apply machine learning and biological data analysis skills at Assembly Biosciences to accelerate antiviral drug discovery through rigorous, insight-driven computational research.

CORE COMPETENCIES

AI/ML Methods | Computational Analysis | Drug Discovery | Feature Engineering | Predictive Modeling | Data Visualization | Biological Data | Python Programming

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built **Random Forest and K-Means models** on 100,000+ fiber-optic network records - including upload speed, latency, and jitter features - to classify network quality across thousands of nodes.
- Developed an end-to-end clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for maintenance planning.
- Reduced manual analysis time by **approximately 60%** through automated feature engineering, model evaluation, and reproducible data processing pipelines built in Python and Pandas.
- Delivered stakeholder presentations translating **model outputs into actionable maintenance recommendations**, communicating complex analytical findings through Power BI dashboards to non-technical audiences.

PROJECTS

Aerial Cactus Detection - Endangered Species Classification

EURECOM Course Project · 2025 · 99.8% Accuracy

- Benchmarked CNN, DenseNet121, and hybrid CNN-Transformer architectures on **17,500+ drone images**, achieving 99.8% accuracy, F1 = 0.999, and ROC AUC = 1.0 for endangered species classification.
- Applied systematic model evaluation and ensemble analysis using **PyTorch and Scikit-learn**, producing reproducible metrics across multiple deep learning architectures for biological image data.

PyTorch · DenseNet121 · Scikit-learn · Pandas · Kaggle GPU

NVIDIA Nemotron Model Reasoning Challenge

Kaggle Competition · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter** Mamba-Transformer with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context).
- Ran **controlled ablation experiments** across learning rate schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers for verified chain-of-thought data.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, Computer Vision, NLP, Image Security, Cloud Computing, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

Data Science & Analytics: Python | SQL | Pandas | NumPy | Scikit-learn | Power BI | Feature Engineering

AI/ML & Deep Learning: PyTorch | Hugging Face Transformers | LoRA / PEFT | Ensemble Methods | Clustering | TensorFlow

MLOps & Visualization: Git | Linux | FP16/BF16 Mixed Precision | Matplotlib | Seaborn | Docker (basic)